Green Protectionism, Carbon Accounting & Market Fragmentation

CHAPTER 2: WEAPONIZATION OF CARBON PRICING & GREEN PROTECTIONISM

2.1 Regulatory Imperialism and the Architecture of the EU CBAM

The European Union's Carbon Border Adjustment Mechanism (CBAM) marks a structural shift in international trade policy. By placing an explicit carbon price on imports based on embedded greenhouse gas (GHG) emissions, CBAM extends internal EU environmental rules to global manufacturing supply chains. While justified under European climate law as a safeguard against "carbon leakage"—where production shifts to countries with looser environmental standards—in practice it functions as a instrument of regulatory power, forcing foreign exporters to align with EU carbon accounting systems or face financial penalties at the border.

The operational mechanics of CBAM center on requiring authorized EU declarants to purchase CBAM certificates. These certificates correspond directly to the weekly average clearing price of European Union Emissions Trading System (EU ETS) allowances, expressed in Euros per metric ton of emitted carbon dioxide equivalent (CO_2e).

CBAM Financial Liability Formula:

$$\text{Net Obligation (€)} = [\ \text{Specific Embedded Emissions } (tCO_2e/t) \times \text{Import Volume (t)}\ ] - [\ \text{Explicit Domestic Carbon Price Paid Paid (€)}\ ]$$

The calculation of embedded emissions requires strict accounting across multiple operational scopes:

- **Scope 1 (Direct Emissions):** GHG emissions produced directly during production processes, including fossil fuel combustion, chemical process reactions, and industrial process heating.
- **Scope 2 (Indirect Emissions from Electricity):** Emissions generated during the production of electricity, heating, or cooling consumed within the manufacturing facility. Scope 2 inclusion is particularly burdensome for developing nations with coal-reliant national grids.
- **Scope 3 (Upstream Precursor Emissions):** Indirect emissions embedded in raw materials and complex input components consumed during production (such as crude steel inputs used in manufacturing downstream specialized machinery).

> **Administrative Enforcement & Default Values Penalty Structure**
> If a foreign installation cannot provide fully verified Scope 1–3 emissions data certified by EU-accredited independent auditors, CBAM rules mandate the application of punitive **Default Values**. These default benchmarks are set at the average emissions intensity of the 10% worst-performing EU installations, maximizing financial liabilities for unverified foreign producers.

2.1 Sectoral Impact Analysis: CBAM Covered Goods

CBAM selectively targets six energy-intensive, trade-exposed (EITE) sectors in its initial enforcement phase. The financial impact varies depending on the carbon intensity of production methods and the energy mix of exporting nations.

Covered Sector	Inclusion Scopes	Primary High-Risk Export Regions	Structural Economic & Technological Disruption
Steel & Iron	Scope 1 & Scope 2	India, China, Turkey, Ukraine, Brazil	Coal-based Blast Furnace-Basic Oxygen Furnace (BF-BOF) production faces steep tariffs compared to European Scrap-EAF or Hydrogen-DRI facilities.
Aluminum	Scope 1 & Scope 2	Mozambique, Bahrain, UAE, India, China	Primary smelting relies heavily on fossil fuel power. High indirect electricity emissions create severe margin compression under Scope 2 integration.
Fertilizers	Scope 1 & Scope 2	Egypt, Algeria, Russia, Trinidad & Tobago	Haber-Bosch production uses natural gas as feedstock and energy source. Nitrous oxide (N_2O) emissions carry high global warming potential multiplier factors.
Cement	Scope 1 Direct	Turkey, Egypt, Ukraine, Vietnam	High process emissions from limestone calcination ($CaCO_3 \rightarrow CaO + CO_2$). Heavy weight-to-value ratio amplifies border price penalties.
Hydrogen	Scope 1 Direct	North Africa, Middle East, Gulf Cooperation Council	Imposes strict origin thresholds; grey/blue hydrogen faces carbon charges, favoring green electrolytic hydrogen produced with certified renewables.
Electricity	Direct Imports	Western Balkans, Ukraine, Turkey	Prevents direct power leakage across land borders; electricity generated from coal-fired grids faces immediate carbon equalization charges.

The administrative burden of CBAM verification disproportionately impacts small and medium-sized enterprises (SMEs) in developing nations. Establishing installation-level monitoring, reporting, and verification (MRV) systems aligned with EU standards requires specialized technical infrastructure and expensive international auditing firms. Foreign firms unable to absorb these compliance costs risk being excluded from European supply chains entirely.

Furthermore, CBAM creates a structural split in global manufacturing. Carbon-intensive producers in the Global South are increasingly forced to divert dirty industrial production to unregulated domestic or non-European markets, while redirecting cleaner production volumes to Europe. This practice—known as "resource shuffling"—avoids border penalties without reducing aggregate global emissions.

2.2 Economic Coercion of Market Access & Revenue Capture

The economic impact of border carbon adjustments extends beyond direct trade charges. It creates macroeconomic disruptions across developing nations that rely heavily on exporting raw materials and basic industrial inputs to the European single market. Because trade revenues from primary commodity exports provide essential foreign currency reserves for many emerging economies, CBAM levies pose risks to exchange rate stability and sovereign debt servicing capacity.

Consider the macroeconomic impact on developing economies with high export concentration in CBAM-covered goods:

- **Export Earnings Degradation:** Border tariffs lower the price competitiveness of carbon-intensive exports, reducing trade volumes and squeezing profit margins for domestic producers.
- **Foreign Exchange Reserve Volatility:** Squeezed export revenues constrain central banks' capacity to accumulate hard currency reserves (USD, EUR), exacerbating local currency depreciation pressures and raising import costs for essential goods.
- **Capital Allocation Skew:** Developing state treasuries must divert scarce capital into high-cost decarbonization projects for export sectors, underfunding domestic climate adaptation, healthcare, and basic infrastructure needs.

> **The Failure of Revenue Recycling Architecture**
> Initial policy proposals for CBAM suggested recycling 100% of border revenues back to the Global South as grant-based climate finance to support industrial decarbonization. However, final EU legislative texts direct these funds into the general EU budget as "Own Resources." Consequently, capital flows directly out of emerging market export industries into Western European treasuries—a transfer of wealth that many Global South leaders view as a form of carbon neo-colonialism.

To avoid border tax leakage to foreign treasuries, several developing economies are moving to establish their own domestic carbon pricing schemes or export taxes. However, implementing an internal carbon price in a low-income nation can weigh on domestic growth, raise energy prices for vulnerable households, and trigger social opposition. Emerging market policymakers face a difficult trade-off: tax their own domestic industries or allow European authorities to collect those revenues at the border.

2.3 Legal Analysis: WTO Disputes & Article XX Exceptions

The rollout of CBAM has created significant tension within international trade law. Developing economies, led by the BASIC group (Brazil, South Africa, India, China), argue that CBAM violates core principles of the World Trade Organization (WTO) and the multilateral climate framework. Specifically, they contend it breaches the principle of Common But Differentiated Responsibilities and Respective Capabilities (CBDR-RC) established under the UNFCCC.

From a strict WTO jurisprudence perspective, CBAM faces serious legal challenges under two primary non-discrimination principles of the General Agreement on Tariffs and Trade (GATT 1994):

- **GATT Article I:1 (Most-Favoured-Nation Treatment):** Mandates that any trade advantage granted to products from one country must be extended immediately and unconditionally to like products from all other WTO members. CBAM's provisions—which grant exemptions or credit adjustments to countries with recognized carbon pricing systems (like Switzerland)—discriminates against nations using non-market regulatory approaches.
- **GATT Article III:4 (National Treatment Principle):** Requires that imported products receive treatment no less favorable than that accorded to like domestic products. Critics argue that while EU producers receive free ETS allowance allocations (being phased out gradually) and electricity price compensation, foreign producers face immediate and stringent compliance burdens.

> **The Article XX General Exceptions Defense**
> The EU defends CBAM's legality under **GATT Article XX**, citing sub-paragraphs **(b)** (measures necessary to protect human, animal, or plant life or health) and **(g)** (measures relating to the conservation of exhaustible natural resources). To survive legal scrutiny, the EU must prove CBAM is not applied in a manner that constitutes a *"means of arbitrary or unjustifiable discrimination"* or a *"disguised restriction on international trade"* under the Article XX Chapeau.

This legal debate is further complicated by the ongoing deadlock in the WTO Appellate Body, caused by blocking judicial appointments. As a result, any formal WTO dispute ruling on CBAM risks being appealed into a legal void, leaving developing nations with little choice but to pursue unilateral trade retaliation.

CHAPTER 3: FRAGMENTATION OF CARBON MARKETS & ARTICLE 6 FAILURES

3.1 The Structural Impasse of Article 6 (Paris Agreement)

Article 6 of the 2015 Paris Agreement was intended to establish a unified, international framework for carbon trading. It enables countries to transfer carbon credits voluntarily to help meet their Nationally Determined Contributions (NDCs). However, negotiations surrounding Article 6 have faced long-standing policy disagreements over baseline accounting, sovereign oversight, and double-counting risks.



A core challenge in Article 6 implementation centers on operationalizing **Corresponding Adjustments (CAs)** to prevent double-counting. When a host nation sells an Internationally Transferred Mitigation Outcome (ITMO) under Article 6.2 to a buyer state or corporate entity, it must add that transferred unit back to its national greenhouse gas inventory.

This creates a trade-off for developing nations. If a host country sells its low-cost carbon abatement options to foreign buyers, it loses those reductions from its own national inventory. Consequently, the country must rely on more expensive domestic measures to meet its own NDC commitments—a phenomenon known as "cherry-picking" high-value mitigation opportunities.

Disagreements over registry transparency, authorized credit usages, and revoked authorization rights continue to delay the deployment of Article 6.4. These delays leave sovereign carbon trading reliant on fragmented bilateral deals under Article 6.2 that often lack uniform accounting standards.

3.2 The Collapse of Trust in Voluntary Carbon Markets (VCMs)

While compliance carbon markets (such as the EU ETS) are established by regulatory mandates, Voluntary Carbon Markets (VCMs) allow corporations and non-state actors to purchase offset credits issued by private registries. Over the past decade, VCMs grew rapidly based on corporate net-zero pledges. However, widespread integrity crises have undermined market confidence, driving credit valuations down across major registries.

Market integrity issues center on four structural failures in offset methodology:

- **Phantom Offsets and Over-crediting:** Empirical studies revealed that leading avoided-deforestation (REDD+) project baselines significantly overestimated forest loss risks, resulting in millions of carbon credits issued for forest protection that would have occurred naturally.
- **Additionality Failures:** Many renewable energy offset projects (such as grid-connected wind and solar farms) were already economically viable without carbon credit revenues, failing the fundamental requirement of environmental additionality.
- **Permanence & Reversal Risks:** Forest and land-based carbon sinks are vulnerable to wildfires, disease, and illegal logging. When stored carbon is re-released into the atmosphere, the underlying offset credit loses its environmental value.
- **North-South Asymmetry & Rent Extraction:** Foreign project developers, intermediaries, and auditing firms often absorb 70–80% of carbon credit sales revenues, leaving local communities and host-nation treasuries with a fraction of the capital generated by domestic resources.

> **Sovereign Reassertion & Export Restrictions**
> In response to low credit valuations and limited local revenue sharing, host governments across the Global South—including Zimbabwe, Kenya, Indonesia, and Honduras—are asserting sovereign control over national carbon assets. These states are introducing national registries, imposing export taxes of up to 50% on carbon sales, or canceling unaligned private offset concessions to protect domestic resources.

3.3 The Rise of Regional and Parallel Carbon Markets

The failure to establish a single global price on carbon under Article 6, combined with growing pushback against Western regulatory frameworks, has led to market fragmentation. Instead of a unified international market, regional and parallel carbon trading platforms are emerging across major non-Western economic blocs.

China is leading this shift by expanding its National Emissions Trading Scheme (ETS). Initially limited to the power generation sector, the Chinese ETS is expanding to cover heavy industrial sectors—including steel, aluminum, and cement. Unlike the EU ETS, which relies on a fixed cap to drive absolute emission reductions, China's ETS uses carbon intensity benchmarks tied to production output. This approach accommodates economic growth while systematically lowering carbon intensity per unit of output.

Concurrently, alternative carbon trading networks are developing across emerging economies:

- **BRICS Carbon Market Platform:** Member states are developing shared carbon accounting standards, mutual credit recognition frameworks, and cross-border settlement mechanisms denominated in local currencies. This architecture bypasses Western verification bodies and insulates member trade from EU CBAM penalties.
- **Bilateral Offset Coalitions:** Programs like Japan's Joint Crediting Mechanism (JCM) bypass complex UN mechanisms by establishing direct bilateral carbon offset agreements with developing partner nations.
- **Regional Voluntary Markets:** High-volume carbon exchanges established in Riyadh (Regional Voluntary Carbon Market Company) and Singapore (CIX) offer localized carbon credit trading tailored to the industrial needs of developing economies.

> **Strategic Impact: Geoeconomic Carbon Bipolarity**
> The global carbon landscape is splitting into two distinct governance spheres: a Western high-cost compliance zone centered on absolute emission caps and border taxes, and a non-Western zone built around intensity benchmarks, sovereign credit control, and flexible bilateral trading. This division fragmenting global supply chains and complicating climate policy coordination.